

CVGlobal and ZeSCO: Geographically Balanced Cross-View Zero-Shot Orientation Estimation



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Code

Context

Objectives

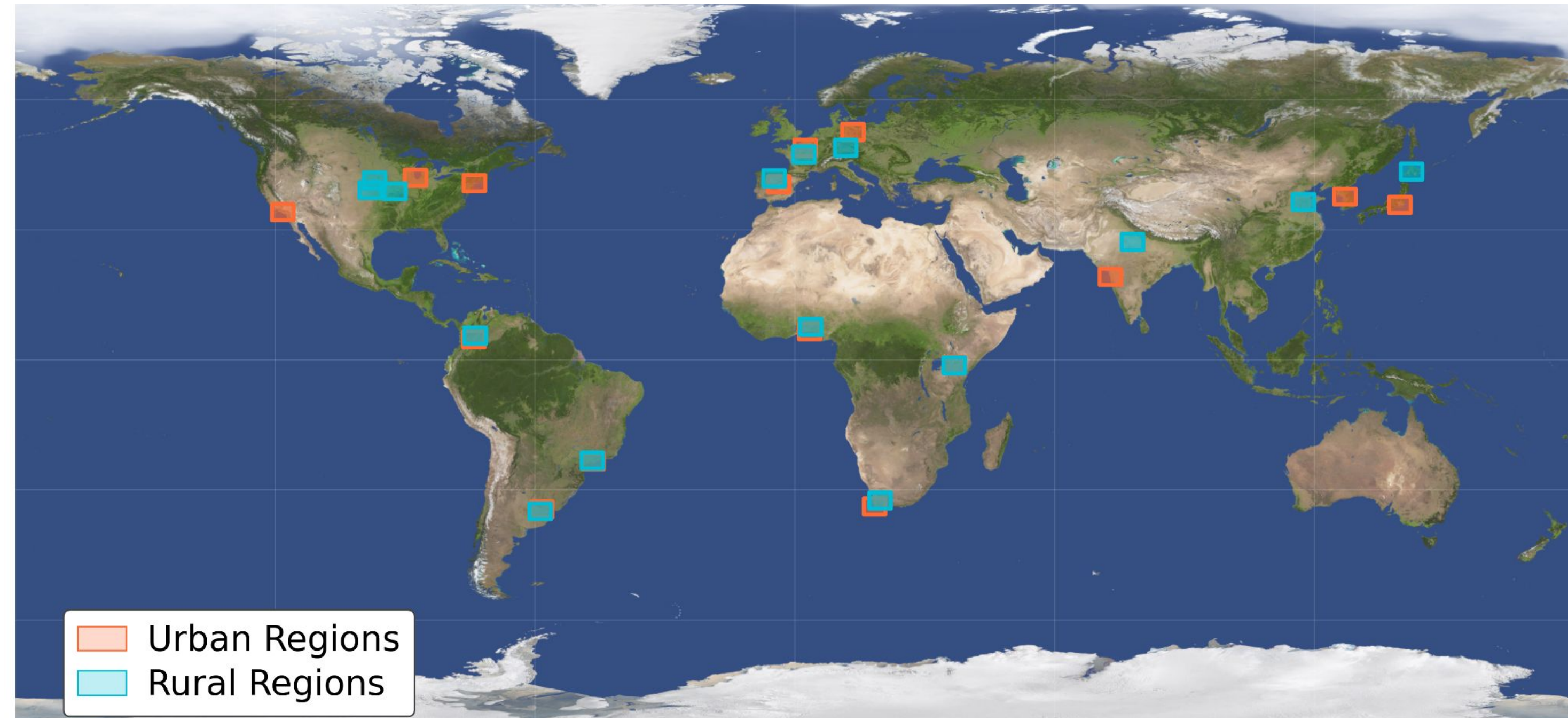
- Estimate relative orientation of ground-level photo w.r.t. co-localized satellite image
- Address geographic bias in existing benchmark datasets (typically single-region) [1]
- Overcome limited generalization of supervised methods on unseen areas

Contributions

1. **Geographically-balanced dataset:** Introduction of **CVGlobal**, a dataset of 10,000 paired images spanning five continents across both urban and rural contexts
2. **Zero-Shot Method:** Introduction of **ZeSCO**, a training-free baseline for cross-view orientation estimation
 - Leverages the cross-modality feature matching capabilities of self-supervised **vision foundation models** (e.g., DINOv3)
 - Integrates lightweight geometric priors and interpretable token aggregation

Proposed dataset

CVGlobal

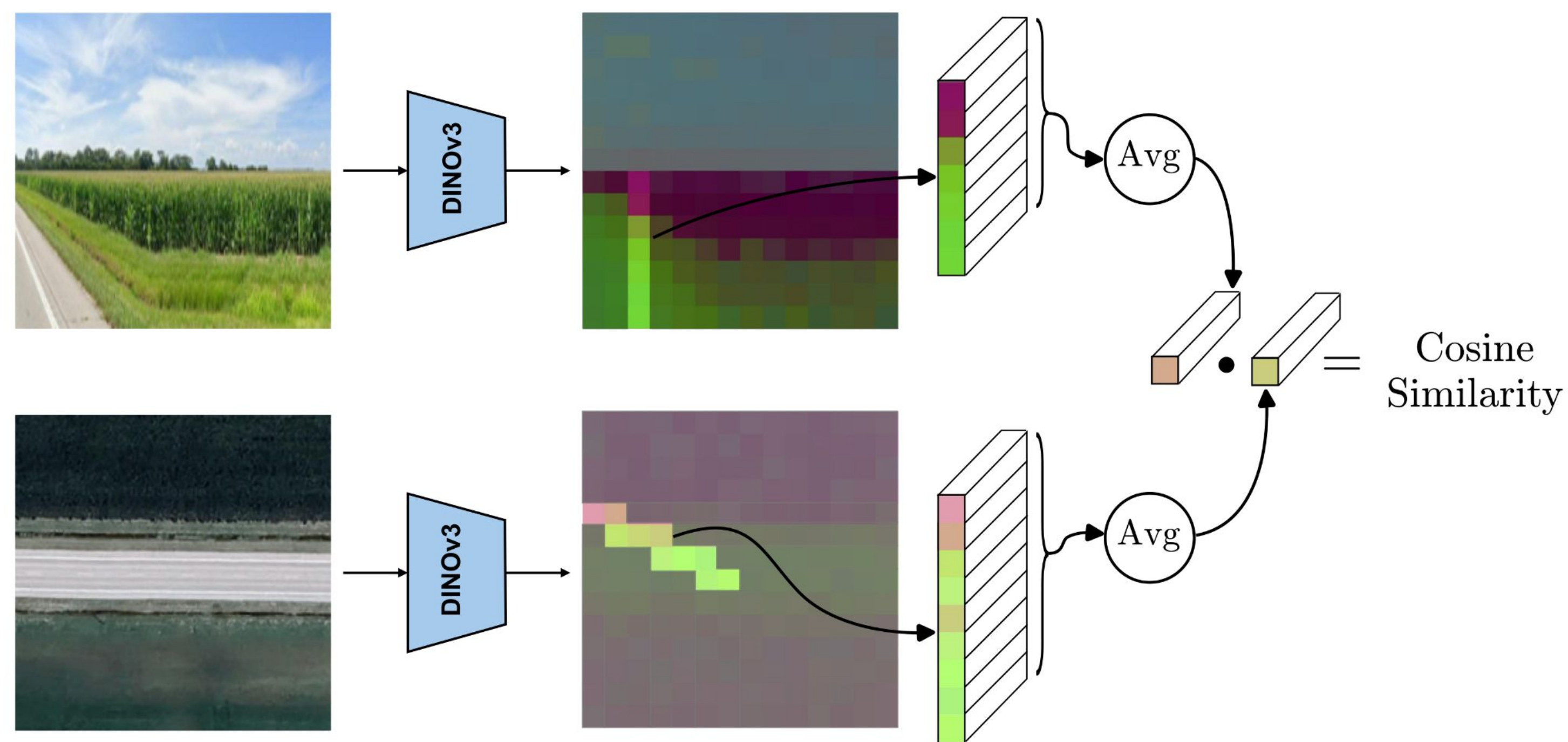


- **Data Source:** Google Maps and Street View Static APIs (640 × 640 px images)
- **Satellite:** GSD ≈ 0.3m/px, 190 x 190 m span
- **Ground:** NWSE directions concatenated

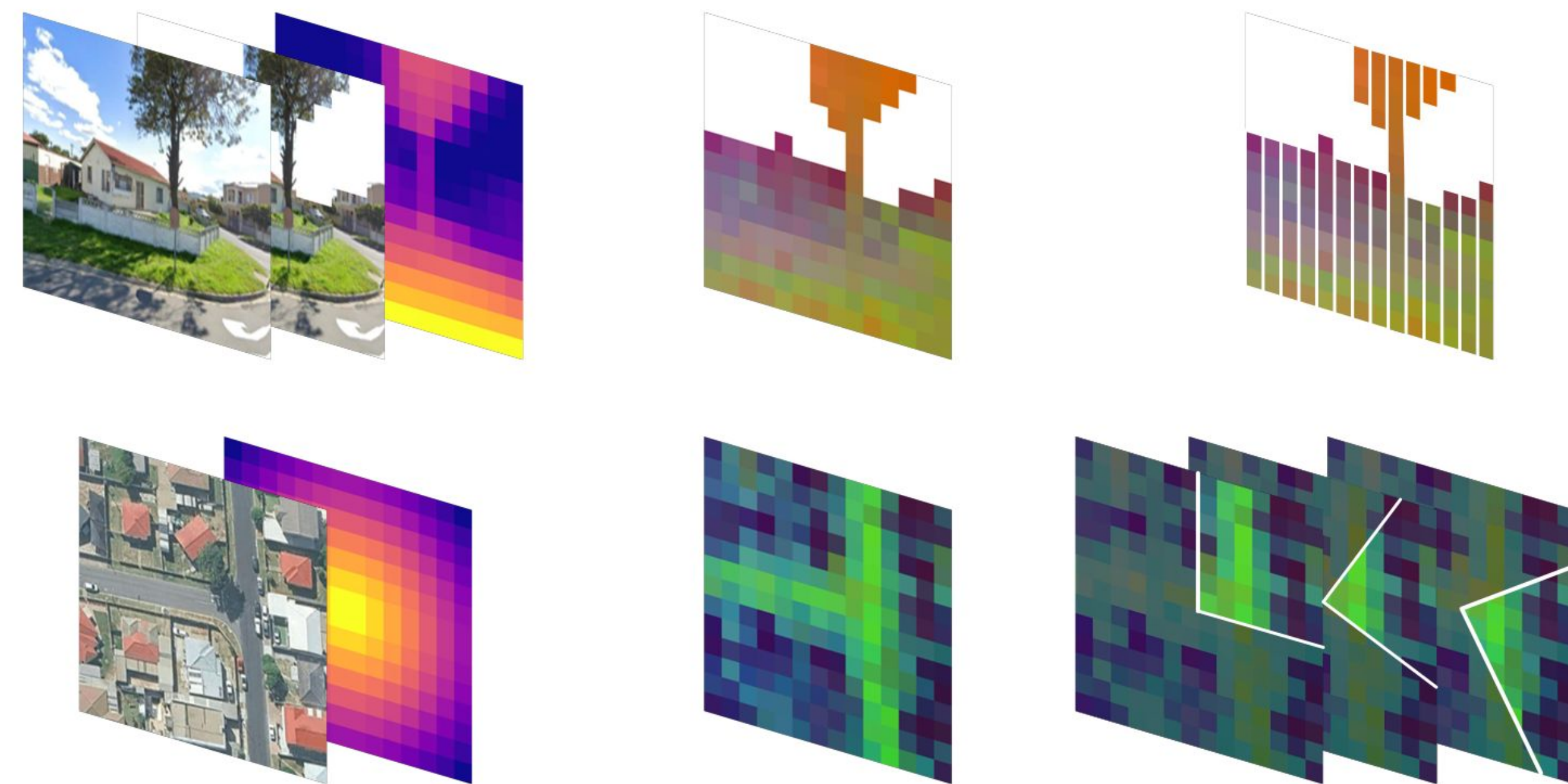
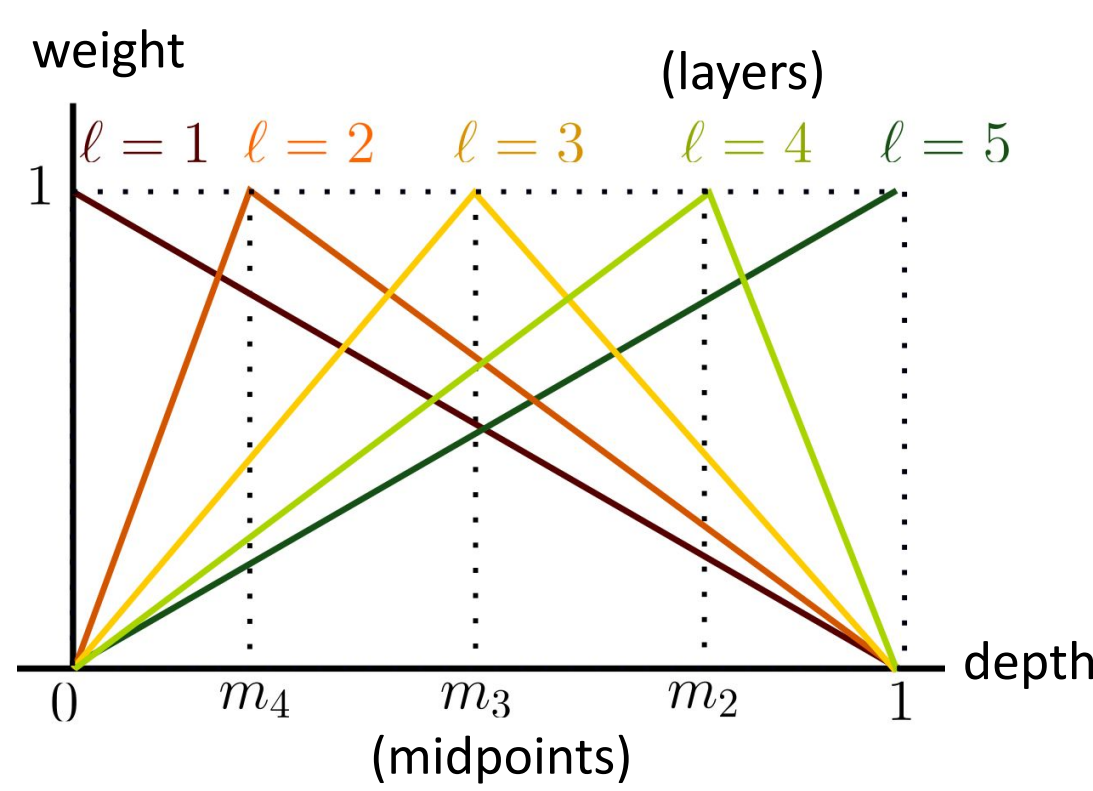
Proposed method

ZeSCO

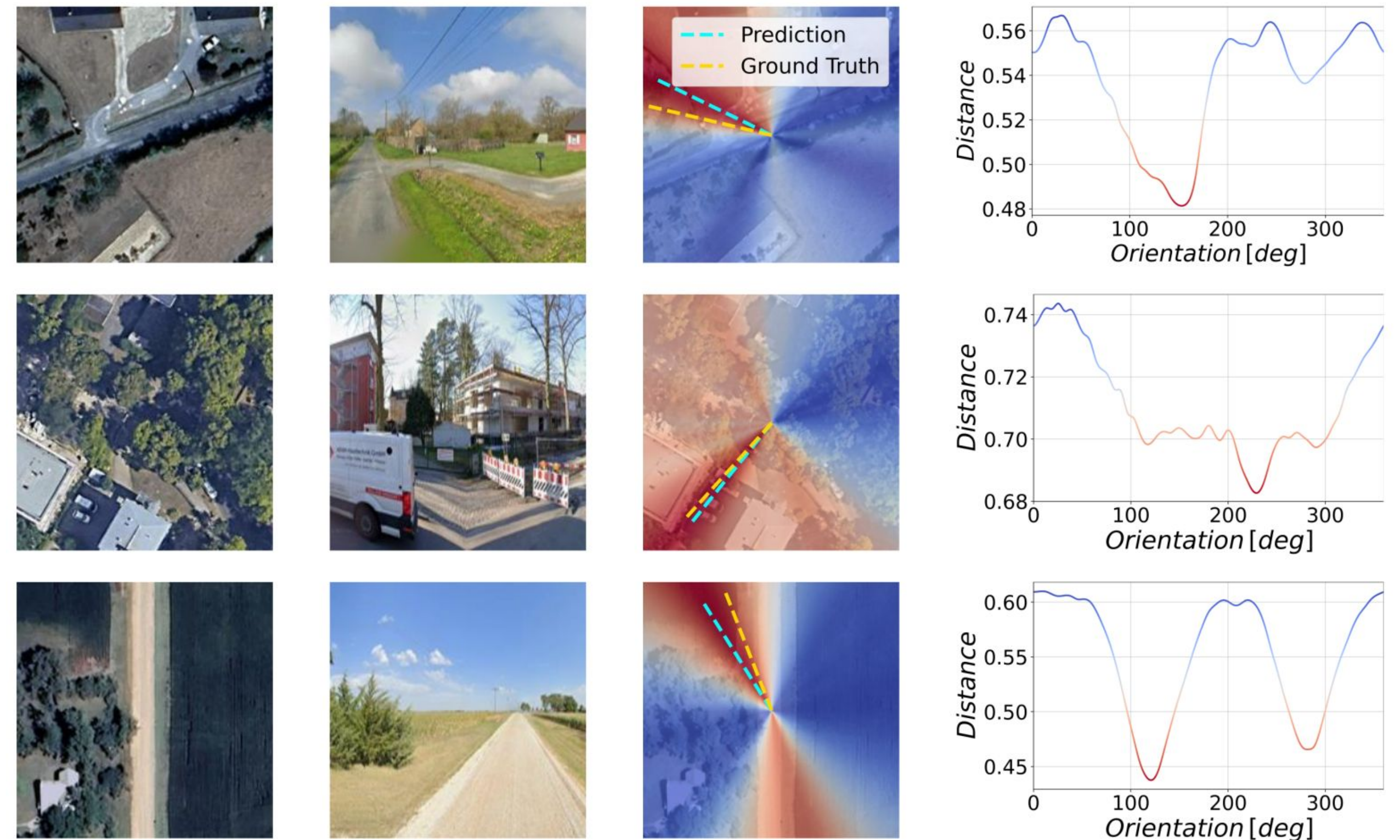
Zero-Shot Cross-View Orientation



1. Feature extraction
 - Pre-trained vision foundation model (DINO, CLIP)
2. Sky filtering [2] and Depth estimation [3]
 - Satellite view: use radial distance from center
3. Depth-Weighted Token Aggregation
4. Orientation estimation via Cosine similarity



Results



Method	Dataset	90°			70°		
		M _{180°} ↓	MAE _{90°} ↓	r@5° ↑	M _{180°} ↓	MAE _{90°} ↓	r@5° ↑
DSM* [4]	CVUSA	4.50°	11.38°	58.91	6.00°	13.54°	48.15
	CVGlobal	11.00°	16.08°	40.94	14.00°	17.65°	31.93
ZeSCO	CVUSA	16.50°	18.60°	22.77	23.50°	21.58°	20.99
	CVGlobal	17.79°	20.32°	19.80	21.50°	22.52°	20.35

Region	Context	DSM*			ZeSCO		
		M _{180°} ↓	MAE _{90°} ↓	r@5° ↑	M _{180°} ↓	MAE _{90°} ↓	r@5° ↑
Africa	Rural	14.00°	17.94°	34.90	21.46°	18.46°	19.00
Africa	Urban	12.00°	16.89°	39.90	14.54°	19.14°	24.50
Asia	Rural	10.00°	15.56°	40.90	21.29°	21.28°	15.00
Asia	Urban	13.00°	16.97°	36.80	27.86°	26.60°	17.00
Europe	Rural	7.00°	14.14°	46.60	16.18°	19.44°	20.00
Europe	Urban	8.50°	14.86°	45.30	13.75°	19.65°	24.50
North America	Rural	7.00°	15.08°	46.80	12.25°	14.28°	27.00
North America	Urban	10.00°	15.35°	43.70	12.82°	18.28°	23.00
South America	Rural	13.00°	17.42°	34.90	30.39°	24.14°	13.50
South America	Urban	12.00°	16.60°	39.60	40.68°	21.92°	14.50

Backbone	M _{180°} ↓	MAE _{90°} ↓	r@5° ↑
CLIP	70.71°	34.03°	9.00
DINOv2	36.56°	24.89°	17.45
DINOv3	17.79°	20.32°	19.80
DINOv3_sat	76.64°	35.84°	9.55

Conclusion and Perspectives

- Advantages of our zero-shot pipeline over supervised ones, despite trailing in accuracy:
 - No task-specific training
 - Easy portability across backbones
 - Compatible with any image FOV
- Drawbacks and limitations:
 - Struggles in highly homogeneous environments
 - Relies on perfectly-centered image pairs
- Future work: integrate light supervision to close the zero-shot gap.

References

- [1] Workman S, Souvenir R, Jacobs N.: Wide-Area Image Geolocalization with Aerial Reference Imagery. In ICCV (2015).
- [2] Leo, L.D., Toffanin, P., Contributors, O.: Skyremoval: Sky mask generation for photogrammetric reconstruction. github.com/OpenDroneMap/SkyRemoval (2022)
- [3] Yang, L., Kang, B., Huang, Z., Xu, X., Feng, J., Zhao, H.: Depth anything: Unleashing the power of large-scale unlabeled data. In CVPR (2024)
- [4] Shi, Y., Yu, X., Campbell, D., Li, H.: Where am i looking at? joint location and orientation estimation by cross-view matching. In CVPR (2020)